

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B. Tech (CBCS) Degree Examinations
 Summer Semester July 2024
EC2001-1 – ANALOG ELECTRONIC CIRCUITS-I
 (EC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- In which region BJT operates when Base emitter junction is forward biased and collector base junction is reverse biased?**
 - Cut-off
 - Active
 - Saturation
 - None of these
- A transistor can be made to operate as a switch by operating it in which of the following regions?**
 - Active Region
 - Active region, Cut off region
 - Active region, Saturation region
 - Saturation region, Cut off region
- What does the negative sign in the voltage gain of the common emitter fixed bias configuration indicate?**
 - Output & input voltages are out of phase by 180°
 - Gain is less than 1
 - Gain is more than 1
 - Output & input voltages are out of phase by 90°
- Operating points of the transistor represents**
 - values of I_C and V_{CE} when signal is applied
 - the magnitude of signal
 - zero signal values of I_C and V_{CE}
 - zero signal values of I_B and V_{BE}
- AC emitter resistance equals 26 mV divided by the**
 - Quiescent base current
 - DC emitter current
 - AC emitter current
 - Change in collector current
- To reduce the distortion in a CE amplifier, reduce the**
 - DC emitter current
 - Base-emitter voltage
 - Collector current
 - AC base voltage
- The output voltage of a CE amplifier is**
 - Amplified
 - Inverted
 - 180° degree out of phase with input
 - All of these
- If the emitter bypass capacitor opens, the AC output voltage will**
 - Decrease
 - Increase
 - Remain the same
 - Equal to zero
- Which of the following terminals does not belong to the MOSFET?**
 - Drain
 - Gate
 - Base
 - Source
- Choose the correct statement**
 - MOSFET is a unipolar, voltage controlled, two terminal device
 - MOSFET is a bipolar, current controlled, three terminal device
 - MOSFET is a unipolar, voltage controlled, three terminal device
 - MOSFET is a bipolar, current controlled, two terminal device
- The output characteristics of a MOSFET, is a plot of**
 - I_d as a function of V_{gs} with V_{ds} as a parameter
 - I_d as a function of V_{ds} with V_{gs} as a parameter
 - I_g as a function of V_{gs} with V_{ds} as a parameter
 - I_g as a function of V_{ds} with V_{gs} as a parameter

12. MOSFET has greatest application in digital circuit due to
 A) Low power consumption B) Less noise
 C) Small amount of space it takes on a chip D) All of the mentioned
13. Which of the following is true about the common Source amplifier with fixed bias?
 A) It has low input impedance B) It has high output impedance
 C) Infinite gain D) Phase reversal voltage output
14. If gain is need to be stabilized, which of the following element is used for a Common source amplifier?
 A) Capacitive B) Inductive
 C) Resistive D) LC tank circuit
15. When gate to source voltage of common source amplifier is at positive peak, drain to source voltage will be
 A) At positive peak B) At negative peak
 C) Infinite D) Zero
16. Which of the following statement is true about MOSFET common source amplifier compared to BJT amplifier?
 A) It has high input impedance B) It has low input impedance
 C) No input voltage is needed D) Input voltage is needed
17. A differential amplifier is capable of amplifying
 A) DC input signal only B) AC input signal only
 C) AC & DC input signal D) None of the mentioned
18. Where does the Q point lie for Class-B amplifier?
 A) Active B) Cut-off
 C) Saturation D) Between saturation and active
19. What is the value of the maximum efficiency of the Class-B amplifier?
 A) 25 % B) 35 %
 C) 35 % to 50 % D) 50 % to 70 %
20. If DC power for a Class-A amplifier is 500 W and AC power is 150 W, what is its efficiency?
 A) 50 % B) 75 %
 C) 20 % D) 30 %

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I

1. a) Explain the operation of fixed-bias configuration with neat circuit diagrams and relevant current and voltage equations.
 b) Derive the expression for input impedance, output impedance, voltage gain for. Emitter follower with relevant diagrams.
2. a) For the emitter-bias network shown in Fig. 2(a), determine the following parameters, (i) I_B (ii) I_C (iii) V_{CE} (iv) V_C (v) V_E (vi) V_B .

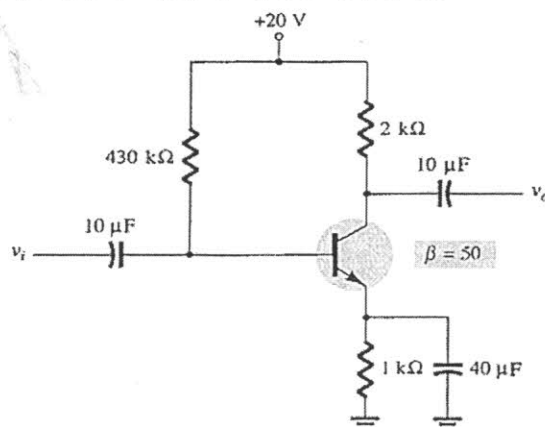


Figure. 2 (a)

Marks BT* CO* PO*

8 L*2 1 1

8 L2 2 1

8 L3 1 1

8 L2 2 1

3. a) Determine the dc bias voltage V_{CE} and the current I_C for the voltage divider configuration of Figure. 3 (a).

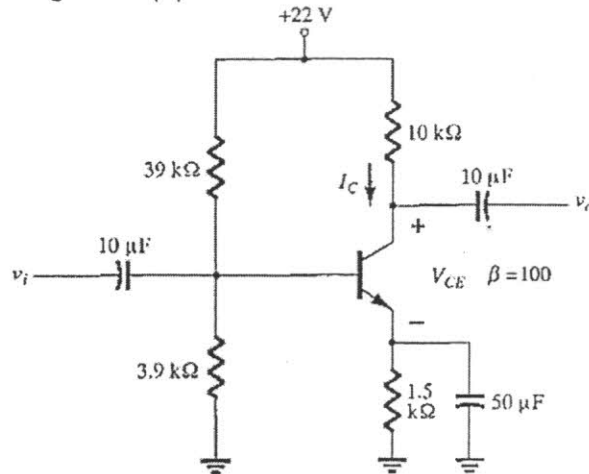


Figure. 3 (a)

- b) For the network of Figure. 3 (b), (i) Determine r_e (ii) Find Z_i (with $r_o = \infty \Omega$) (iii) Calculate Z_o (with $r_o = \infty \Omega$) (iv) Determine A_v (with $r_o = \infty \Omega$)

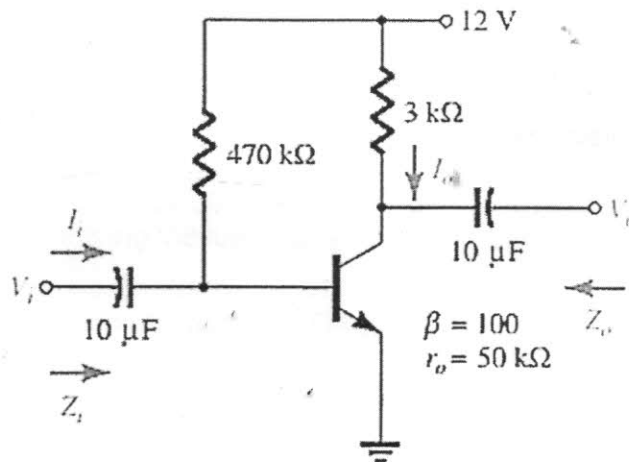


Figure. 3 (b)

Unit – II

4. a) With a neat cross-sectional view, explain the physical structure of the enhancement-type NMOS transistor.
b) Derive the expression for input impedance output impedance, voltage gain of common source amplifier without degeneration. Assume $\lambda = 0$.
5. a) With a neat sketch, explain how the channel is formed between source and drain with the application of gate voltage in enhancement-type NMOS transistor.
b) Derive the expression for input impedance output impedance voltage gain MOS source follower. Assume $\lambda \neq 0$.
6. a) With the help of volt-ampere characteristic, explain the operating regions of enhancement-type NMOS transistor.

- b) Calculate the small-signal voltage gain of the CS stage shown in Figure 6(b) if $I_D = 1\text{mA}$, $\mu_n C_{ox} = 100\mu\text{A/V}^2$, $V_{Th} = 0.5\text{V}$ and $\lambda = 0$. Verify that M_1 operates in saturation.

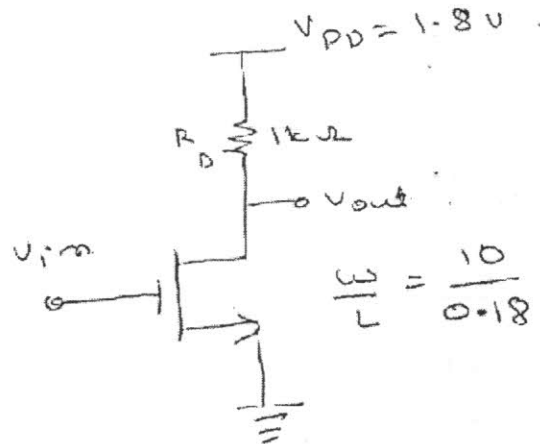


Figure. 6 (b)

8 L3 4 1

Unit – III

7. a) With a neat circuit diagram and relevant equations, explain MOS transistor based current-steering circuit. 8 L2 5 1
 b) Derive the expression of power-conversion efficiency for class-A amplifier. 8 L3 5 1
8. a) With a neat amplifier circuit diagram and transfer characteristic, explain the operation of Class-B output stage. 8 L2 5 1
 b) Design a Class-B output stage to deliver an average power of 20 W to an 8 Ω load. The power supply is to be selected such that V_{CC} is about 5 V greater than the peak output voltage. 8 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
